

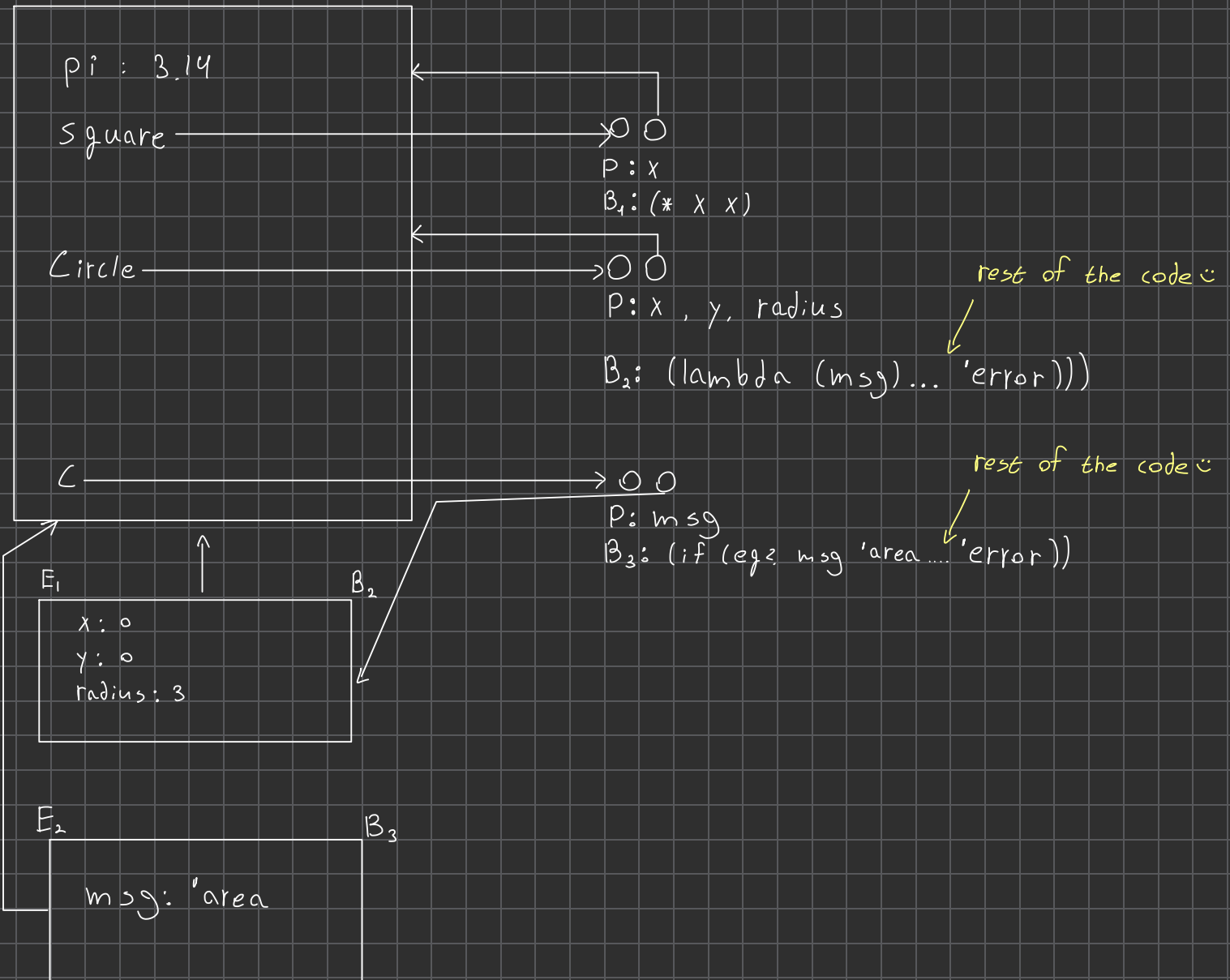
```

1 ;1. Convert the ClassExps to ProcExps
2 ; Before Conversion
3 (define pi 3.14)
4 (define square (lambda (x) (* x x)))
5 (define circle
6   (class (x y radius)
7     (
8       (area (lambda () (* (square radius) pi)))
9       (perimeter (lambda () (* 2 pi radius)))
10    )
11  )
12 )
13 (define c (circle 0 0 3))
14 (c 'area)
15
16
17 ;After Conversion
18 (define pi 3.14)
19 (define square (lambda (x) (* x x)))
20 (define circle
21   (lambda (x y radius)
22     (lambda (msg)
23       (if (eq? msg 'area)
24         (lambda () (* (square radius) pi))
25         (if (eq? msg 'perimeter)
26           (lambda () (* 2 pi radius))
27           'error)
28       )
29     )
30   )
31 )
32 )
33 (define c (circle 0 0 3))
34 (c 'area)
35
36
37 ;2. List the expressions which are passed to the L3applicativeEval function during the
38 ; computation of the program, (after the conversion), for the case of substitution
39 model.
40 ;1
41 3.14
42
43 ;2
44 (lambda (x) (* x x))
45
46
47 ;3
48 (lambda (x y radius)
49   (lambda (msg)
50     (if (eq? msg 'area)
51       (lambda () (* (square radius) pi))
52       (if (eq? msg 'perimeter)
53         (lambda () (* 2 pi radius))
54         'error)
55     )
56   )
57 )
58 )
59
60
61 ;4
62 (circle 0 0 3)
63
64 ;5
65 circle
66
67 ;6
68 0

```

```
69
70 ;7
71 0
72
73 ;8
74 3
75
76 ;9
77 (lambda (msg)
78   (if (eq? msg 'area)
79       (lambda () (* (square 3) pi))
80       (if (eq? msg 'perimeter)
81           (lambda () (* 2 pi 3))
82           'error
83       )
84   )
85 )
86
87 ;10
88 (c 'area)
89
90 ;11
91 c
92
93 ; 12
94 'area
95
96 ;13
97 (if (eq? 'area 'area)
98     (lambda () (* (square 3) pi))
99     (if (eq? 'area 'perimeter)
100         (lambda () (* 2 pi 3))
101         'error
102     )
103 )
104
105 ;14
106 (eq? 'area 'area)
107
108 ;15
109 eq?
110
111 ;16
112 'area
113
114 ;17
115 'area
116
117 ;18
118 (lambda () (* (square 3) pi))
119
120
121
```

GE



Value: (lambda () (* (square radius) pi))